Fiscal Year:	FY 2009	**Task Last Updated:**	FY 08/27/2009
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Project Title:	Effects of Space Radiation on Degenerative Tissue Disease, Genetic Instability and Oxidative DNA Damage in Ataxia Telangiectasis Deficient Mice		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation Biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	(1) **Cardiovascular**:Risk of Cardiovascular Adaptations Contributing to Adverse Mission Performance and Health Outcomes		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	90095-8347	**Congressional District:**	33
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2004 Radiation Biology NNH04ZUU005N
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No. of Post Docs:	1	**No. of PhD Degrees:**	1
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA ARC
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Flight Program:			
Flight Assignment:	NOTE: Received NCE to 8/14/2010 per A. Chu/ARC (8/09) NOTE: Changed Division and Discipline/Program to HRP as of FY2006, per program changes then, per JSC/A. Chu-ARC (jvp 4/2009)		
Key Personnel Changes/Previous PI:	N/A		
COI Name (Institution):	Fleming, Sheila (University of California, Los Angeles)		
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Performance Goal No.:			
Performance Goal Text:			

Task Description:	Atm (ataxia telangiectasia mutated) heterozygosity has a very high prevalence in the human population and seems to be a risk factor for cardiac damage and cancer. Radiation also causes reactive oxygen species, which also seem to be involved in the etiology of Ataxia Telangiectasia, cardiac damage, movement disorder and cancer. Oxidative DNA damage, DNA strand breaks and genetic instability may be involved in the etiology of these diseases. Atm deficient homozygous patients and mice show an elevated frequency of DNA deletions and signs of oxidative stress in their cells. Preliminary results show that DNA deletions at the pun locus are induced by ionizing radiation. The DNA deletion assay is based on the quantification of black spots on fur and eyes resulting from reversion of the pun mutation. This reversion occurs by deletion of 70 kb of an internal duplication within the p gene. Aim 1 is to determine whether AtmDSRI homozygous or heterozygous mice may show an increased frequency of space radiation induced DNA deletions. We also found that the elevated frequency of DNA deletions in Atm truncation deficient mice is completely reverted to wildtype levels by nutritional supplementation with the radical scavenger and glutathione precursor N-acetyl cysteine (NAC). Thus, we will test in aim 2 whether AtmDSRI homozygous or heterozygous mice nutritionally supplemented with NAC may show reduced frequencies of space radiation induced DNA deletions whereas mice supplemented with the prooxidant and inhibitor of glutathione metabolism, BSO may show an increased frequency of radiation induced deletions, which will be tested in aim 2. In aim 3 we will test whether AtmDSRI homozygous and heterozygous mice may show an increased frequency of space radiation induced cardiac damage or cancer. Aim 4 is to test whether AtmDSRI homozygous and heterozygous mice may show an increased frequency of space radiation induced ataxia. Implications for human health may include that Atm missense mutation heterozygosity (a significant proportion of the human population) may cause an increased risk for space radiation induced cardiac damage ataxia or cancer and that potential astronauts may benefit from being tested for Atm heterozygosity and that chemoprevention with antioxidants may reduce the space radiation induced genomic instability and possibly cancer in Atm heterozygous people.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Implications for human health may include that Atm missense mutation heterozygosity (a significant proportion of the human population) may cause an increased risk for space radiation induced cardiac damage, ataxia or cancer and that potential astronauts may benefit from being tested for Atm heterozygosity and that chemoprevention with antioxidants may reduce the space radiation induced genomic instability and possibly cancer in Atm heterozygous people.
Task Progress:	ATM is a key gene involved in the cellular response to ionizing radiation induced DNA damage. The frequency of heterozygous ATM mutant carriers in the United States population may be as high as 1%. Our research seeks to investigate whether mice heterozygous mutant for Atm exhibit increased sensitivity to high LET space radiation induced cancer, neurological, and/or cardiovascular diseases as well as other long-term effects of radiation including fibrosis and inflammation. Furthermore we would like to look at differences in DNA damage caused by high LET compared to low LET radiation in wildtype, Atm heterozygous and Atm mutant mice. Our fourth year primarily consisted of collecting data on the long-term effects of high LET (1 or 2 Gy, 1 GeV 56Fe) radiation on Atm mutant, heterozygous, and wildtype mice. Specifically, we looked at lifespan, data from necropsy reports, and neuromotor function. At this time, all mice in our study have reached our experimental endpoint of 2 years and have been archived for future studies. 1 Gy, 1 GeV 56Fe ions affects lifespan in a sex and genotype-specific manner. We found that high LET radiation does not affect the survival of Atm-SRI homozygous or heterozygous mice. Atm mutant mice, however, have a decreased lifespan compared to Atm-SRI heterozygous and wildtype mice regardless of radiation ($p<0.001$). Atm-SRI heterozygous mice also have a decreased lifespan compared to wildtype mice without radiation ($p<0.05$). Atm-SRI wildtype mice, however, have decreased lifespan when exposed to 1 Gy, 1 GeV/n 56Fe ions ($p<0.05$). The median lifespan for non-irradiated and irradiated wt mice is 102.1 and 93 weeks, respectively. We also looked at sex differences between exposed and non-exposed mice. We found that female Atm-SRI homozygous female mice spontaneously have a shorter median lifespan than their male counterparts (42.4 vs 63.3 weeks, $p<0.005$), but this difference disappears if mice are exposed to 1 Gy, 1 GeV 56Fe ions. Atm-SRI heterozygous and wildtype female mice, however do not have a significantly different lifespan than their male counterparts spontaneously, but may be more sensitive to 1 Gy, 1 GeV 56Fe ions in terms of lifespan. Necropsies on mice which have died or have been euthanized are incomplete, and so far do not reveal many effects of radiation, but do show some differences among genotypes. We have looked at diseases or pathologies such as cancer, solid tumors, cardiovascular-related diseases, fibrosis, inflammation, cataracts, and extramedullary hematopoiesis. We have completed necropsy reports of about 80% of our total mice. We have sent at least 25 mice of each genotype with and without exposure to radiation for complete necropsies. So far we see that spontaneously, Atm-SRI homozygous mice have an increased frequency of neoplastic formation compared to wildtype mice, however radiation does not affect carcinogenesis and/or tumorigenesis regardless of genotype. 1 Gy, 1 GeV 56Fe ions, however seems to affect the formation of solid tumors in Atm-SRI heterozygous mice, which have a low spontaneous frequency of solid tumors that increases when exposed to HZE radiation. When we looked at frequencies of mice with cancer, we did not see a difference between irradiated and non-irradiated mice, however it is possible that high LET radiation decreases latency of disease rather than influences incidence. 1 Gy, 1 GeV 56Fe ions shifts the survival curve of Atm-SRI heterozygous mice to the left, or decreases the age at which Atm-SRI heterozygous mice develop cancer and/or tumors ($p<0.05$) while wildtype mice are not affected. Oddly, exposure of Atm-SRI homozygous mice to HZE radiation shifts the curve of cancer latency to the right ($p<0.01$). We hope to examine possible causes of this shift in the upcoming year. To look at neuromotor ability in mice we used three different tests. We tested gait consistency, spontaneous activity, and balance. To measure gait consistency, hind paws were painted, and their stride length and range were measured. To measure spontaneous activity, mice were put in a cylinder, and activities such as rearing or grooming were analyzed. To measure balance, mice traversed an elevated 1m long beam which decreases in width. Mice were trained for 2 days, then video-taped for analysis. For each test we used at least 7 mice per group. So far we have analyzed footprint data which measures step length and gait variability (by looking at the range between the shortest and longest step). We found that 2 Gy, 1 GeV 56Fe ions affects gait variability in all genotypes. Neither exposure to HZE radiation nor genotype affected spontaneous activity. We are still waiting for data analysis on most of the balance data.

Bibliography Type:	Description: (Last Updated: 08/25/2020)
Abstracts for Journals and Proceedings	Yamamoto ML, Reliene R, Hacke K, Hafer K, Westbrook A, Bral M, Trouiller B, Schiestl RH. "Effects of high LET radiation on DNA damage, neuromotor function, carcinogenesis and longevity in Ataxia telangiectasia deficient mice." Heavy Ions in Therapy and Space Symposium, Cologne, Germany, July 6-10, 2009. Meeting proceedings. Heavy Ions in Therapy and Space Symposium, Cologne, Germany, July 6-10, 2009. , Jul-2009
Dissertations and Theses	Hafer K. "Biological Protection against Low- and High-LET Ionizing Radiation." Dissertation, University of California, Los Angeles, August 2008. , Aug-2008